



Modelling Financial Markets

Chapter 4: Portfolio Optimization



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ ERC: Equal Risk Contribution portfolio
- ▣ ETF: Exchange-Traded Fund
- ▣ FDR: False Discovery Rate
- ▣ FWER: Family-Wise Error Rate
- ▣ GMV: Global Minimum-Variance portfolio
- ▣ GMV-LO: Global Minimum-Variance portfolio, Long-Only
- ▣ GMV-LW: Global Minimum-Variance portfolio with Ledoit-Wolf covariance
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HRP: Hierarchical Risk Parity
- ▣ HYG: iShares iBoxx High Yield Corporate Bond ETF
- ▣ IEF: iShares 7–10 Year Treasury Bond ETF
- ▣ LO: Long-Only, no short sales
- ▣ LQD: iShares iBoxx Investment Grade Corporate Bond ETF
- ▣ MV: Mean–Variance tangency portfolio estimated from sample moments
- ▣ MV-LO: Mean–Variance tangency portfolio, Long-Only
- ▣ OOS: Out-Of-Sample
- ▣ US: United States

Outline

Motivation

Mean–Variance Optimisation

The Limits of Markowitz

Covariance Shrinkage

Black–Litterman

Risk Parity and Hierarchical Risk Parity

Out-of-Sample Evaluation

A Portfolio of BVB Blue Chips

AI for Scientific Discovery

Conclusions

References

What Is Portfolio Optimisation?

Definition

- ▣ A **portfolio** is a vector of weights $w = (w_1, \dots, w_N)^\top$, the shares of wealth invested in N assets, with $w^\top \mathbf{1} = 1$
- ▣ **Portfolio optimisation**: choose w to maximise a criterion (return per unit of risk, utility) subject to constraints

Key message of this chapter

- ▣ The optimal portfolio is easy to write and hard to estimate: judge every rule **out of sample**, after costs, with a test

Why It Matters

- ▣ **Asset managers and pension funds:** strategic allocation across equities, bonds, credit, gold
- ▣ **Private investors:** how much in local stocks, how much abroad?
- ▣ **Risk management:** the same covariance matrix drives portfolio risk (Chapters 6–7)
- ▣ **Econometrics:** N means and $N(N + 1)/2$ covariances estimated from short samples
- ▣ Textbook treatment: [Franke, Härdle & Hafner \(2019\)](#), Part I (portfolio theory)

The Central Tension

In theory

- ▣ Known means and covariances
- ▣ The mean–variance portfolio is optimal
- ▣ More assets always help

In practice

- ▣ Parameters are estimated with error
- ▣ The optimiser loads on the largest errors
- ▣ More assets, more parameters, more noise

- ▣ This chapter: better inputs (shrinkage, Black–Litterman), rules that ignore means (risk parity, HRP) and a fair out-of-sample comparison

Data Used in This Chapter

Universe	Assets	Months
Sectors	9 US sector ETFs (XLB ... XLY)	1999–2026
Multi-asset	SPY, EFA, EEM, TLT, IEF, LQD, HYG, GLD	2007–2026
Combined	9 sectors + 7 multi-asset ETFs (no SPY)	2007–2026
Romania	8 BVB blue chips, BET, BET-TR	2014–2026

- Monthly returns from month-end prices on common trading days; US data end in July 2026, BVB data in August 2026
- Risk-free rate: one-month Treasury bill, [Kenneth French Data Library](#)

The Assets

- ▣ Equities: SPY (S&P 500), EFA (developed markets outside North America), EEM (emerging markets)
- ▣ US Treasuries: TLT (maturities above 20 years), IEF (7–10 years)
- ▣ Credit: LQD (investment-grade corporate bonds), HYG (high-yield corporate bonds)
- ▣ Gold: GLD; sectors: XLK technology, XLU utilities, XLP consumer staples, . . .
- ▣ BVB: TLV, SNP, BRD, TGN, SNG, SNN, EL, TEL; BET is a price index, BET-TR reinvests dividends

Notation and Performance Measures

- Simple return $R_{i,t} = P_{i,t}/P_{i,t-1} - 1$; excess return $R_{i,t}^e = R_{i,t} - R_{f,t}$
- Portfolio: $R_{p,t} = \mathbf{w}^\top \mathbf{R}_t$, mean $\mu_p = \mathbf{w}^\top \boldsymbol{\mu}$, variance $\sigma_p^2 = \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}$
- Annualisation of monthly figures: mean $\times 12$, volatility $\times \sqrt{12}$
- **Sharpe ratio** ([Sharpe \(1966\)](#)): excess return per unit of volatility

$$SR_p = \frac{\mathbb{E}[R_p^e]}{\sigma(R_p^e)}$$

- **Maximum drawdown**: largest fall of wealth from a previous peak, $\min_t (W_t / \max_{s \leq t} W_s - 1)$

The Markowitz Problem

- ▣ **Markowitz (1952)**: investors care about the mean and the variance of portfolio returns
- ▣ **Efficient frontier**: minimum variance for every target mean m

$$\min_w w^\top \Sigma w \quad \text{s.t.} \quad w^\top \mu = m, \quad w^\top \mathbf{1} = 1$$

- ▣ Equivalent utility form, risk aversion γ : $\max_w w^\top \mu - \frac{\gamma}{2} w^\top \Sigma w$
- ▣ Inputs: N means, N variances, $N(N-1)/2$ covariances
- ▣ Weights may be negative: a **short sale** (selling a borrowed asset)

Two Assets: Diversification

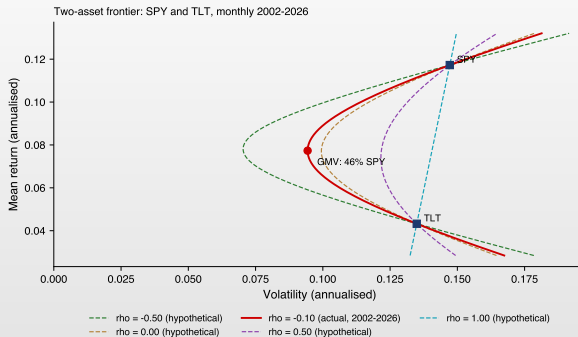
- Weight w in asset 1, $1 - w$ in asset 2, correlation ρ

$$\sigma_p^2 = w^2 \sigma_1^2 + (1 - w)^2 \sigma_2^2 + 2w(1 - w)\rho \sigma_1 \sigma_2$$

- $\rho = 1$: no diversification, σ_p is linear in w
- $\rho < 1$: σ_p lies below the straight line; $\rho = -1$: a riskless mix exists
- Minimum-variance weight (set $\partial \sigma_p^2 / \partial w = 0$)

$$w^* = \frac{\sigma_2^2 - \rho \sigma_1 \sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho \sigma_1 \sigma_2}$$

Two-Asset Frontiers: Stocks and Treasuries



□ SPY and TLT, monthly total returns, August 2002 – July 2026 ($T = 288$)

Reading the Two-Asset Frontier

- ▣ SPY: mean 11.7%, volatility 14.7%; TLT: mean 4.3%, volatility 13.5%
- ▣ Sample correlation $\rho = -0.10$: Treasuries hedged equities on average
- ▣ Minimum-variance mix: 46% SPY, volatility 9.4%, lower than either asset
- ▣ The curves for hypothetical ρ show that correlation, not volatility, drives diversification
- ▣ Caveat: the correlation changes over time: daily -0.40 in 2002–2021, $+0.08$ in 2022 (Chapter 6)

Worked Example: The Minimum-Variance Mix, Step by Step

- ▣ Inputs: SPY $\sigma_1 = 0.1472$, TLT $\sigma_2 = 0.1349$, $\rho = -0.1005$
- ▣ Numerator: $\sigma_2^2 - \rho\sigma_1\sigma_2 = 0.01820 + 0.00199 = 0.02019$
- ▣ Denominator: $\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2 = 0.02166 + 0.01820 + 0.00399 = 0.04385$
- ▣ $w^* = 0.02019/0.04385 = 0.461$ in SPY, 0.539 in TLT
- ▣ $\sigma_p = 9.4\%$, mean 7.7%: two-thirds of the SPY mean with less than two-thirds of its risk

Tobin: Separation



James Tobin, Nobel Prize in Economics 1981

Photo: unknown author (1962); public domain;
[Wikimedia Commons](#)

- **Tobin (1958)**: add a riskless asset paying R_f
- Every investor holds the **same** risky portfolio, mixed with cash or borrowing
- Risk aversion changes only the mix, not the risky portfolio: **two-fund separation**
- The best risky portfolio maximises the Sharpe ratio: the **tangency portfolio**
- Separation is the bridge from Markowitz to the CAPM (Chapter 3)

Two Special Portfolios

Global minimum variance (GMV)

- Lowest variance of all portfolios

$$w_{\text{GMV}} = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}$$

- Needs **no** expected returns

- In this chapter, MV denotes the tangency rule estimated from sample means and covariances

Tangency (maximum Sharpe)

- Highest Sharpe ratio

$$w_{\text{MV}} = \frac{\Sigma^{-1} \mu}{\mathbf{1}^\top \Sigma^{-1} \mu}$$

- μ : expected **excess** returns

Merton: the Frontier in Closed Form



Robert C. Merton, Nobel Prize in
Economics 1997

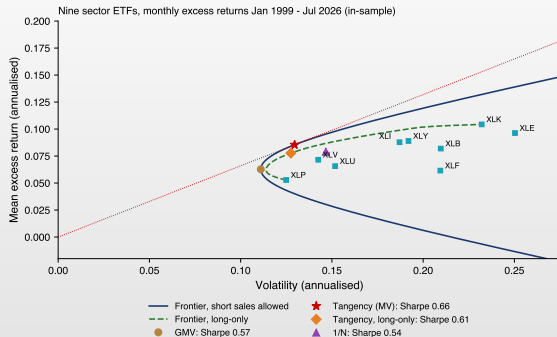
Photo: Digarnick (2006); public domain;
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- Merton (1972): with short sales, the frontier is a hyperbola
- Let $A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$, $B = \mathbf{1}^\top \Sigma^{-1} \boldsymbol{\mu}$, $C = \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu}$, $D = AC - B^2$

$$\sigma^2(m) = \frac{Am^2 - 2Bm + C}{D}$$

- The GMV sits at $m = B/A$ with $\sigma^2 = 1/A$
- Every frontier portfolio is a mix of GMV and tangency

The Frontier of US Sectors



- Monthly excess returns, January 1999 – July 2026 ($T = 331$); all parameters estimated on the full sample

Reading the Sector Frontier

- ▣ Tangency: Sharpe 0.66; GMV: 0.57 (volatility 11.1%); $1/N$: 0.54
- ▣ Tangency weights: -0.37 financials, -0.21 materials, $+0.37$ health care
- ▣ Long-only tangency: Sharpe 0.61; materials, financials and industrials get exactly zero
 - ▶ A **corner solution**: the constraint binds and assets drop out
- ▣ The gain of the optimiser over $1/N$ is small even **in sample**, with perfect hindsight

Constraints

- ▣ **Long-only (LO)**: $w_i \geq 0$; no closed form, solved by quadratic programming
- ▣ Position limits: $w_i \leq 0.2$; sector or asset-class limits: $\sum_{i \in G} w_i \leq c_G$
- ▣ Leverage limit: $\sum_i |w_i| \leq L$
- ▣ **Frost & Savarino (1988)**: constraining weights improved out-of-sample performance
- ▣ In this chapter: MV-LO and GMV-LO are the long-only versions of MV and GMV

Recap: Mean–Variance Optimisation

- ▣ The frontier trades mean against variance; correlation drives diversification
- ▣ Two special portfolios: GMV, which needs no means, and tangency, which maximises the Sharpe ratio
- ▣ With a riskless asset, the risky part of every efficient portfolio is the tangency portfolio
- ▣ Long-only constraints create corner solutions
- ▣ Everything so far assumed the parameters are **known**

Means Are Hard to Estimate

- Standard error of an annualised mean over Y years: $\sigma/\sqrt{Y}$, whatever the sampling frequency
- Merton (1980): sampling more often sharpens variances, not means
- Covariances: the estimation error falls with the **number of observations**
- Chopra & Ziemba (1993): errors in means cost about ten times more than errors in variances
- Hence rules that need no means (GMV, risk parity) are often more robust

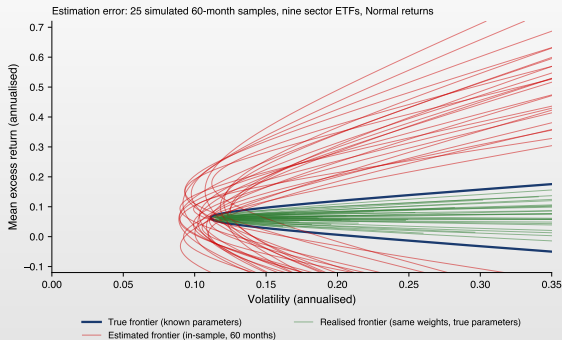
Worked Example: How Many Years for a Precise Mean?

- SPY, 24 years of monthly data: mean 11.7%, volatility 14.7%
- Standard error: $0.147/\sqrt{24} = 3.0\%$; 95% interval [5.8%, 17.6%]
- For a standard error of 1%: $Y = (0.147/0.01)^2 \approx 217$ years
- An optimiser treats a 3% estimation error as information
- Monthly or daily data do not help: only the length of the sample does

Error Maximisation

- Michaud (1989): mean–variance optimisers are **estimation-error maximisers**
 - ▶ They overweight assets with overestimated means and underestimated risk
- Best & Grauer (1991): tiny changes in a mean can move the weights of many assets
- In-sample performance is therefore **biased upwards**
- Experiment: simulate returns from known parameters, optimise, evaluate with the true parameters

Estimated versus Realised Frontiers



- True parameters: sector sample means and covariances, 1999–2026; returns drawn from the Normal distribution

Reading the Simulation

- Red: frontiers estimated from 60 months, far too optimistic
- Green: the same portfolios evaluated with the true parameters, all **below** the true frontier
- 2000 simulated tangency portfolios, 60 months each
 - ▶ Median in-sample Sharpe 1.57; median true Sharpe 0.30; true optimum 0.66
 - ▶ 96% have a true Sharpe below $1/N$ (0.54); 7.5% below zero
- Even with Normal, stable returns, estimation error destroys the optimum

Unstable Weights



- Nine sector ETFs; each month the weights use only the previous 60 months (top panel clipped at ± 6)

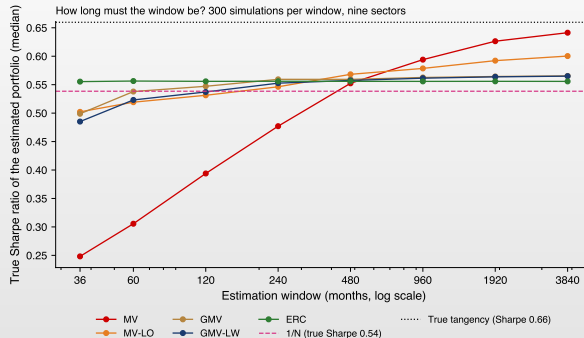
Reading the Rolling Weights

- MV: median gross exposure $\sum_i |w_i| = 4.9$; largest single weight 26 in absolute value
- Gross exposure above 3 in 90% of the months
- Explosions in 2004–2006 and 2009: $1^\top \Sigma^{-1} \hat{\mu}$ close to zero in the denominator
- GMV with shrinkage (next section): gross exposure 1.5, largest weight 0.94
- Unstable weights mean high turnover, hence high trading costs

The 1/N Benchmark

- ▣ **1/N**: equal weights $w_i = 1/N$, rebalanced each month; no estimation at all
- ▣ DeMiguel, Garlappi & Uppal (2009): 14 optimisation models on 7 data sets
 - ▶ None beats 1/N consistently in Sharpe ratio, certainty-equivalent return or turnover
- ▣ Their calibration: about 3000 months of data for sample MV to beat 1/N with 25 assets
- ▣ Kan & Zhou (2007): combining tangency and GMV reduces estimation risk
- ▣ Question: how long must the window be for our nine sectors?

How Long Must the Window Be?



- 300 simulated samples per window length; true parameters as in the previous simulation

Reading the Window-Length Experiment

- ▣ MV beats 1/N only with 480 months (40 years) of data: median true Sharpe 0.55 vs 0.54
- ▣ With 60 months, MV reaches 0.31, less than 60% of 1/N
- ▣ Long-only MV, GMV and GMV with shrinkage are close to 1/N from 60 months on
- ▣ ERC (Section 6) is almost insensitive to the window: 0.56 throughout
- ▣ Here 1/N is nearly efficient: the true tangency Sharpe is only 0.66

Recap: The Limits of Markowitz

- ▣ Means are estimated with large errors; the optimiser amplifies them
- ▣ In-sample Sharpe ratios of optimised portfolios are biased upwards
- ▣ Sample MV weights are unstable and extreme
- ▣ $1/N$ is a hard benchmark: MV needs decades of data to beat it
- ▣ Remedies: better covariance estimates, better means, or rules that avoid means

Why the Sample Covariance Matrix Fails

- ▣ N assets: $N(N + 1)/2$ variances and covariances; 16 assets: 136 parameters from 60 months
- ▣ Sample eigenvalues are too dispersed: the largest too large, the smallest too small
- ▣ **Condition number**: largest over smallest eigenvalue; a large value means Σ^{-1} amplifies noise
- ▣ Optimisers invert Σ , so they load on the directions with the smallest estimated variance
- ▣ If $T < N$, the sample matrix is singular and cannot be inverted

The Shrinkage Estimator

- Shrink the sample matrix S towards a structured target F

$$\hat{\Sigma}_{\text{LW}} = \delta F + (1 - \delta) S, \quad 0 \leq \delta \leq 1$$

- Targets

- ▶ Single-index (market) model: Ledoit & Wolf (2003)
- ▶ Scaled identity: Ledoit & Wolf (2004a)
- ▶ Constant correlation, $f_{ij} = \bar{r}\sqrt{s_{ii}s_{jj}}$: Ledoit & Wolf (2004b), used here

- δ trades the variance of S against the bias of F

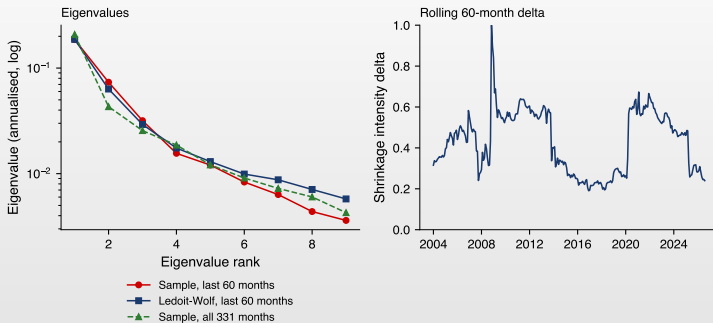
The Optimal Shrinkage Intensity

- Ledoit–Wolf choose δ to minimise the expected squared (Frobenius) distance to the true Σ

$$\delta^* = \max \left\{ 0, \min \left\{ 1, \frac{\hat{\kappa}}{T} \right\} \right\}, \quad \hat{\kappa} = \frac{\hat{\pi} - \hat{\rho}}{\hat{\gamma}}$$

- $\hat{\pi}$: sum of the asymptotic variances of the entries of S (noise)
- $\hat{\rho}$: their covariances with the target; $\hat{\gamma} = \|F - S\|^2$ (misspecification)
- Noisy data or a good target: large δ ; long samples: $\delta \rightarrow 0$

Shrinkage on US Sectors



- Nine sector ETFs, monthly excess returns; last window August 2021 – July 2026

Reading the Shrinkage Chart

- ▣ Last window: $\delta = 0.24$; the smallest eigenvalues are pulled up, the largest almost unchanged
- ▣ Condition number: sample 51.6, shrunk 32.5
- ▣ Rolling δ : mean 0.43, range 0.19–1.00
 - ▶ $\delta = 1$ in the early windows: 60 months are too few to beat the target
- ▣ GMV-LW: the GMV portfolio computed with $\hat{\Sigma}_{LW}$

Constraints Are Shrinkage Too

- Jagannathan & Ma (2003): the long-only GMV equals the unconstrained GMV of a modified covariance matrix
 - ▶ The constraint shrinks the covariances of the assets it binds on
- Imposing the “wrong” constraint (no short sales) can reduce out-of-sample risk
- The effect is largest when covariances are estimated from few observations
- Practical rule: shrink, constrain, or both, but never invert a raw sample matrix of many assets

Recap: Covariance Shrinkage

- The sample covariance matrix is noisy and ill-conditioned when N is large relative to T
- Shrinkage mixes it with a structured target; the data choose the weight δ
- The target matters: constant correlation suits one asset class better than a mix of stocks and bonds
- No-short constraints act as implicit shrinkage

Black and Litterman: Starting from Equilibrium



Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia; public domain; Wikimedia Commons

- Black & Litterman (1992), developed at Goldman Sachs
- Problem: sample means give extreme, unintuitive weights
- Idea: start from the returns that make a **benchmark** portfolio optimal
- Add the investor's **views**, weighted by their confidence
- Result: weights that move away from the benchmark only where views exist

Step 1: Reverse Optimisation

- A mean–variance investor with risk aversion δ_{RA} holds $w^* = (\delta_{RA}\Sigma)^{-1}\mu$
- Reverse it: the returns that make the benchmark w_b optimal

$$\pi = \delta_{RA} \Sigma w_b$$

- π : **implied** (equilibrium) excess returns; with market weights, the CAPM returns
- Here: no market capitalisations for the sectors, so $w_b = 1/N$, $\delta_{RA} = 2.5$

Step 2: Views and the Posterior

▣ K views: $P\boldsymbol{\mu} = \mathbf{q} + \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim N(0, \boldsymbol{\Omega})$; prior $\boldsymbol{\mu} \sim N(\boldsymbol{\pi}, \tau\boldsymbol{\Sigma})$

▣ Posterior mean: a precision-weighted average

$$\boldsymbol{\mu}_{\text{BL}} = [(\tau\boldsymbol{\Sigma})^{-1} + P^\top \boldsymbol{\Omega}^{-1} P]^{-1} [(\tau\boldsymbol{\Sigma})^{-1} \boldsymbol{\pi} + P^\top \boldsymbol{\Omega}^{-1} \mathbf{q}]$$

▣ Weights: $w_{\text{BL}} = (\delta_{\text{RA}} \boldsymbol{\Sigma})^{-1} \boldsymbol{\mu}_{\text{BL}}$

▣ He & Litterman (2002): $\tau = 0.05$, $\boldsymbol{\Omega} = \text{diag}(P \tau \boldsymbol{\Sigma} P^\top)$; then only the assets in the views move

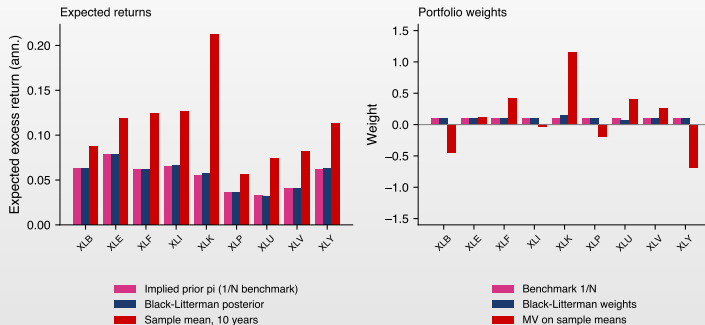
Worked Example: One Asset, One View

- Prior $\pi = 5\%$, $\sigma = 16\%$, $\tau = 0.05$: prior variance $\tau\sigma^2 = 0.00128$
- View $q = 8\%$ with standard deviation 2% : $\omega = 0.0004$
- Precisions: $1/0.00128 = 781$, $1/0.0004 = 2500$

$$\mu_{\text{BL}} = \frac{781 \times 5\% + 2500 \times 8\%}{781 + 2500} = 7.29\%$$

- The view gets weight 0.76 because it is more precise than the prior
- Halve the confidence (standard deviation 4%) and the weight falls to 0.44

Black-Litterman on US Sectors



□ Σ from monthly excess returns, August 2016 – July 2026; one view: XLK minus XLU = 3% a year

Reading Black–Litterman

- ▣ Implied XLK minus XLU: 2.2%; view 3%; posterior 2.6% (the average, since Ω equals the prior variance of the view)
- ▣ Weights: XLK from 11.1% to 14.4%, XLU from 11.1% to 7.8%, all others unchanged
- ▣ Sample means instead: XLK 21.3% a year; MV weights +115% XLK, −68% XLY, −46% XLB
- ▣ Black–Litterman is Bayesian shrinkage of the means towards equilibrium

Recap: Black–Litterman

- The prior comes from a benchmark, not from noisy sample means
- Views enter with explicit confidence; weights move only where views exist
- Results depend on the benchmark, τ , Ω and the risk aversion
- Without views, Black–Litterman returns the benchmark itself

Risk Contributions

- ▣ Volatility is homogeneous of degree one in the weights (Euler)

$$\sigma_p = \sum_{i=1}^N w_i \frac{\partial \sigma_p}{\partial w_i} = \sum_{i=1}^N \frac{w_i (\Sigma w)_i}{\sigma_p}$$

- ▣ **Risk contribution** of asset i : $RC_i = w_i (\Sigma w)_i / \sigma_p$; percentage share RC_i / σ_p
- ▣ Equal weights are not equal risks: volatile assets dominate
- ▣ Risk budgeting: choose w so that the shares RC_i / σ_p equal preset budgets b_i

Equal Risk Contribution

- ERC (Maillard, Roncalli & Teïletche (2010)): $RC_i = RC_j$ for all i, j
- Computed from a convex problem, then normalised to sum to one

$$\min_{y>0} \frac{1}{2} y^T \Sigma y - \frac{1}{N} \sum_i \ln y_i, \quad w = y / 1^T y$$

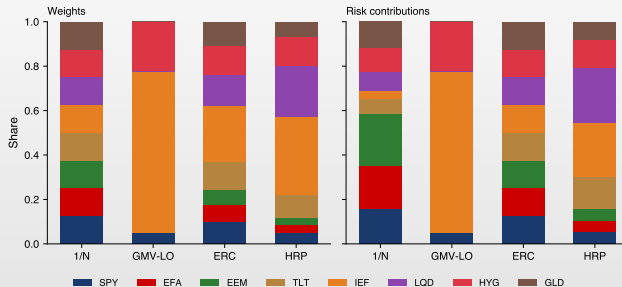
- Volatility ordering: $\sigma_{\text{GMV}} \leq \sigma_{\text{ERC}} \leq \sigma_{1/N}$
- Two assets: ERC is **inverse volatility**, $w_1 = \sigma_1^{-1} / (\sigma_1^{-1} + \sigma_2^{-1})$, for any ρ
- Needs no expected returns; only Σ

Worked Example: ERC with Stocks and Treasuries

- ▣ SPY volatility 14.7%, IEF volatility 6.6% (monthly, 2002–2026), $\rho = -0.10$
- ▣ ERC weight in SPY: $\frac{1/0.147}{1/0.147+1/0.066} = 0.31$
- ▣ Check: $w_1(\Sigma w)_1 = w_2(\Sigma w)_2$, each half of the variance
- ▣ A 60/40 portfolio instead: SPY carries 94% of the risk
- ▣ Risk parity funds lever the ERC portfolio to reach a target volatility

Weights versus Risk Contributions

Eight multi-asset ETFs, monthly May 2007 - Jul 2026



- Eight multi-asset ETFs, full-sample covariance matrix, May 2007 – July 2026 ($T = 231$)

Reading the Risk Contributions

- ▣ 1/N: equities and high yield receive half of the weight but 69% of the risk
- ▣ ERC: 12.5% of the risk from every asset; IEF gets the largest weight (25%)
- ▣ Long-only GMV: 73% IEF, 22% HYG, 5% SPY; five assets at zero
- ▣ Volatility: 1/N 9.4%, ERC 7.9%, HRP 7.1%, long-only GMV 5.8%
- ▣ ERC lies between 1/N and GMV, as the theory predicts

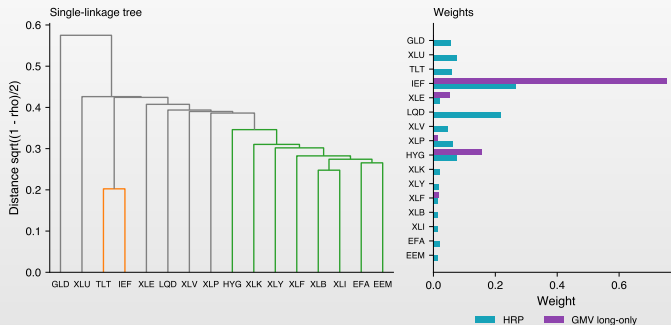
Hierarchical Clustering of Assets

- ▣ Correlation distance: $d_{ij} = \sqrt{(1 - \rho_{ij})/2} \in [0, 1]$
- ▣ **Single linkage**: repeatedly merge the two closest clusters; cluster distance = closest pair
- ▣ The result is a tree (**dendrogram**); cutting it gives groups of similar assets
- ▣ Mantegna (1999): the same distance and the **minimum spanning tree** reveal the hierarchy of a stock market
- ▣ Textbook: Härdle & Simar (2019), cluster analysis

Hierarchical Risk Parity

- HRP (López de Prado (2016)), three steps
 - ▶ Tree clustering on d_{ij} (single linkage)
 - ▶ Quasi-diagonalisation: reorder assets so that similar ones are adjacent
 - ▶ Recursive bisection: split the ordered list in two; split the weight in inverse proportion to the variance of each half
- Variance of a half: inverse-variance weights inside it, $\tilde{w}^\top \Sigma \tilde{w}$
- No matrix inversion: works even when Σ is singular
- All weights are positive

HRP on 16 ETFs



- Nine sectors and seven multi-asset ETFs, monthly, May 2007 – July 2026

Reading the Tree

- ▣ Closest pair: TLT and IEF (distance 0.20); gold joins last (0.57)
- ▣ Equity cluster: cyclical sectors, EFA, EEM and high-yield credit
- ▣ Defensive sectors (utilities, staples, health care) and energy join the equity cluster late
- ▣ HRP: all 16 weights positive, largest IEF 26.8% and LQD 21.9%
- ▣ Long-only GMV: 75.6% in IEF and only five assets above zero

Other Risk-Based Rules

- ▣ **Inverse volatility:** $w_i \propto 1/\sigma_i$; equals ERC when all correlations are equal
- ▣ **Maximum diversification** (Choueifaty & Coignard (2008)): maximise $w^\top \sigma / \sqrt{w^\top \Sigma w}$
- ▣ **Minimum variance:** GMV, the extreme risk-based rule
- ▣ All ignore expected returns: robust to estimation error, but optimal only under special assumptions
 - ▶ ERC is the tangency portfolio if all assets have the same Sharpe ratio and equal correlations

Recap: Risk-Based Allocation

- ▣ Risk contributions decompose portfolio volatility additively
- ▣ ERC equalises them; it lies between $1/N$ and GMV
- ▣ HRP uses a correlation tree and needs no matrix inversion
- ▣ These rules avoid expected returns entirely: less estimation error, no view on returns

Backtest Design

- ▣ **Out-of-sample** (OOS): weights at month t use only data up to $t - 1$
- ▣ Rolling window of 60 months; rebalancing every month
- ▣ Eight rules: 1/N, MV, MV-LO, GMV, GMV-LO, GMV-LW, ERC, HRP
- ▣ Three universes: sectors (OOS January 2004 – July 2026, 271 months), multi-asset and combined (May 2012 – July 2026, 171 months)
- ▣ Performance of excess returns: annualised mean, volatility, Sharpe ratio, maximum drawdown

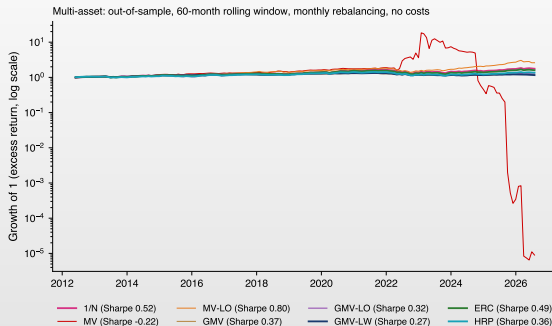
Turnover and Transaction Costs

- Between rebalancings weights drift with returns

$$w_{i,t}^+ = \frac{w_{i,t}(1 + R_{i,t})}{\sum_j w_{j,t}(1 + R_{j,t})}$$

- Turnover:** traded fraction of wealth, $TO_{t+1} = \sum_i |w_{i,t+1} - w_{i,t}^+|$
- Net return with cost c per unit traded: $R_{p,t}^{\text{net}} = R_{p,t} - c TO_t$
- 1/N also trades: it sells winners and buys losers every month

Out-of-Sample Growth: Multi-Asset



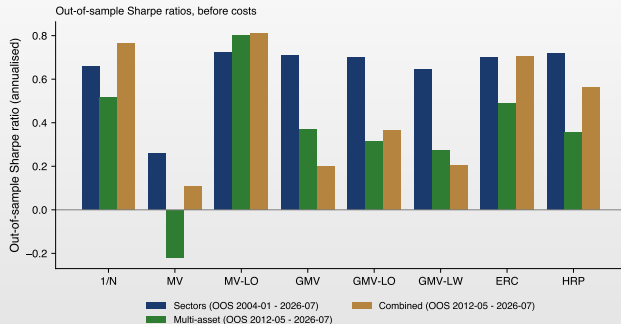
- Eight multi-asset ETFs; wealth of 1 invested in the excess returns, before costs

MFM_ch4_oos_backtest

Reading the Multi-Asset Backtest

- ▣ MV collapses: annualised volatility 518%, turnover 44 times wealth per month
- ▣ MV-LO has the highest Sharpe (0.80); 1/N 0.52; ERC 0.49
- ▣ GMV rules have the lowest volatility (3.8%–5.4%) but also the lowest Sharpe (0.27–0.37)
 - ▶ They load on Treasuries, whose excess returns were slightly negative after May 2012 (TLT -0.5% , IEF -0.2% a year)
- ▣ Is MV-LO really better, or lucky? A test is needed

Sharpe Ratios Across Universes



60-month rolling window, monthly rebalancing, annualised, before costs

MFM_ch4_oos_backtest

Results: Sector Universe

Rule	Mean	Volatility	Sharpe	Turnover	Max. drawdown
1/N	9.4%	14.3%	0.66	0.02	−50%
MV	17.8%	69.0%	0.26	3.86	−100%
MV-LO	10.7%	14.7%	0.72	0.23	−44%
GMV	8.5%	11.9%	0.71	0.23	−34%
GMV-LO	8.1%	11.5%	0.70	0.09	−36%
GMV-LW	7.6%	11.8%	0.65	0.14	−32%
ERC	9.3%	13.2%	0.70	0.03	−48%
HRP	9.0%	12.5%	0.72	0.08	−45%

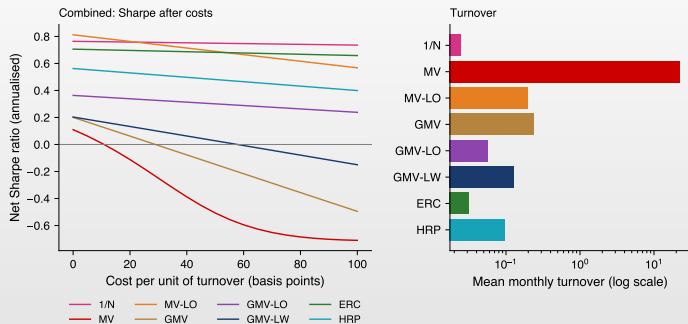
- January 2004 – July 2026; turnover: mean monthly traded fraction of wealth
- All rules except MV land within 0.07 of 1/N

Results: Combined Universe (16 ETFs)

Rule	Mean	Volatility	Sharpe	Turnover	Sharpe, 50 bp
1/N	7.9%	10.3%	0.76	0.02	0.75
MV	28.7%	261.8%	0.11	22.10	-0.51
MV-LO	8.0%	9.8%	0.81	0.20	0.69
GMV	0.8%	4.1%	0.20	0.24	-0.15
GMV-LO	2.0%	5.4%	0.36	0.06	0.30
GMV-LW	0.9%	4.4%	0.20	0.13	0.03
ERC	5.5%	7.8%	0.71	0.03	0.68
HRP	3.9%	7.0%	0.56	0.10	0.48

- May 2012 – July 2026; last column: net of 50 basis points per unit of turnover
- 1 basis point (bp) = 0.01%

Sharpe Ratios after Costs



- Combined universe (16 ETFs); cost c from 0 to 100 bp per unit of turnover

Reading the Cost Chart

- ▣ Turnover ranges from 0.02 (1/N) to 22 (MV) per month
- ▣ Net Sharpe turns negative at about 11 bp for MV, 29 bp for GMV, 58 bp for GMV-LW
- ▣ MV-LO overtakes 1/N only for costs below about 22 bp (Seminar, B8)
- ▣ 1/N and ERC barely move: low turnover is a hidden advantage
- ▣ Rule: report net performance for realistic costs, not only gross

Testing a Difference in Sharpe Ratios

- H_0 : $SR_1 = SR_2$ for two strategies evaluated on the same months
- Jobson & Korkie (1981): a test under i.i.d. Normal returns
- Ledoit & Wolf (2008): a robust version
 - ▶ $\hat{\Delta} = f(\hat{v})$, $v = (\mu_1, \mu_2, \mathbb{E}[r_1^2], \mathbb{E}[r_2^2])$, $f = \frac{\mu_1}{\sqrt{\mathbb{E}r_1^2 - \mu_1^2}} - \frac{\mu_2}{\sqrt{\mathbb{E}r_2^2 - \mu_2^2}}$
 - ▶ Standard error by the delta method: $\sqrt{\nabla f^\top \hat{\Psi} \nabla f / T}$, $\hat{\Psi}$ a HAC covariance
- Strategies on the same assets are highly correlated: small differences can be significant

The Block Bootstrap

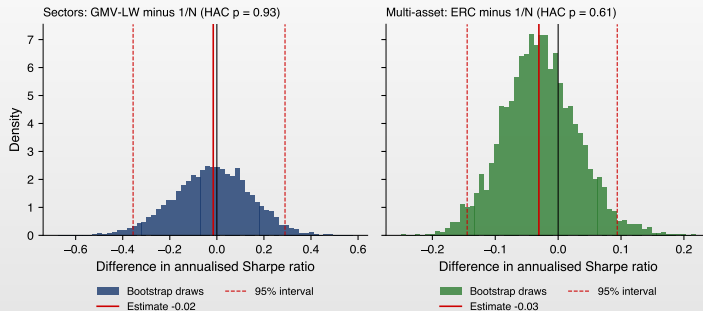


Bradley Efron, inventor of the bootstrap
(1979)

Photo: Ryan K. Morris (2007); public
domain; Wikimedia Commons

- Bootstrap: resample the data to approximate the sampling distribution of $\hat{\Delta}$
- Returns are dependent (volatility clustering): resample **blocks** of consecutive months (Künsch (1989))
- Circular version (as in Ledoit & Wolf (2008)): blocks wrap around the end of the sample
- Here: blocks of 6 months, 5000 draws; pairs of strategy returns resampled together
- p -value: share of draws with $|\hat{\Delta}^* - \hat{\Delta}| \geq |\hat{\Delta}|$

Is Anything Better than $1/N$?



- Bootstrap distributions of the Sharpe ratio difference against $1/N$, annualised

Reading the Tests

- ▣ Sectors, GMV-LW minus 1/N: -0.02 , 95% interval $[-0.36, 0.29]$, $p = 0.93$
- ▣ Multi-asset, ERC minus 1/N: -0.03 , interval $[-0.15, 0.09]$, $p = 0.59$
 - ▶ ERC and 1/N are so correlated that the interval is narrow, yet it still covers zero
- ▣ Sectors: no rule differs from 1/N at 5%; the closest is ERC, $+0.04$ ($p = 0.09$)
- ▣ Multi-asset: MV-LO $+0.29$ is **not** significant ($p = 0.12$)

Significant Differences: Mostly Losses

Universe	Rule minus 1/N	$\hat{\Delta}$	HAC p	Bootstrap p
Multi-asset	MV	-0.74	0.01	0.06
Multi-asset	HRP	-0.16	0.07	0.07
Combined	GMV	-0.57	0.03	0.03
Combined	GMV-LO	-0.40	0.05	0.05
Combined	GMV-LW	-0.56	0.02	0.03
Combined	MV-LO	+0.05	0.82	0.83

- No rule beats 1/N significantly in any universe
- GMV rules lose significantly on the combined universe: low risk, but too little return
- 21 tests: a few p -values near 0.05 are expected by chance (Chapter 3, multiple testing)

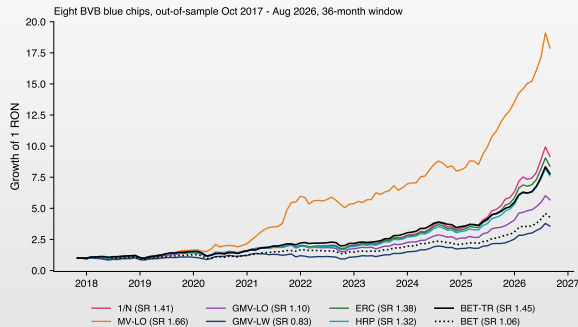
Recap: Out-of-Sample Evaluation

- Only out-of-sample, net-of-cost results are informative
- Sample MV fails in every universe; constraints and shrinkage tame it
- Risk-based rules and $1/N$ are hard to separate statistically
- Test Sharpe differences with HAC errors or a block bootstrap, and count your tests

Setup: a Romanian Investor

- ▣ Eight blue chips listed since 2014: TLV, SNP, BRD, TGN, SNG, SNN, EL, TEL
- ▣ Monthly returns in RON, October 2014 – August 2026; 36-month window, first portfolio October 2017
- ▣ Rules: 1/N, MV-LO, GMV-LO, GMV-LW, ERC, HRP; benchmarks BET-TR and BET
- ▣ Sharpe ratios with $R_f = 0$ (no RON money-market series in the data): return per unit of risk, higher than excess-return Sharpe ratios
- ▣ Earlier Romanian evidence on diversification: Dragotă, Șerbănescu & Pele (2008)

BVB Portfolios versus BET-TR



□ Out-of-sample, October 2017 – August 2026 (107 months), before costs

Reading the BVB Backtest

- ▣ BET-TR: mean 24.7% a year, volatility 17.0%, Sharpe 1.45; BET: mean 17.8%, Sharpe 1.07
 - ▶ The gap of about 7% a year is the dividends: compare with BET-TR, never with BET
- ▣ 1/N 1.41, ERC 1.38, HRP 1.32: close to BET-TR
- ▣ MV-LO 1.66 (difference +0.21, $p = 0.44$); GMV-LW 0.83 (-0.63 , $p = 0.05$)
- ▣ Average correlation of the eight stocks: 0.44; the latest GMV-LO puts 70% in TLV

Beyond Equities: Digital Assets

- ▣ [Petukhina, Trimborn, Härdle & Elendner \(2021\)](#): cryptocurrencies in portfolio allocation rules; gains depend on the rule and on liquidity
- ▣ [Pele et al. \(2023\)](#): are cryptos becoming alternative assets?
- ▣ Bitcoin: annualised volatility 172% (monthly, 2010–2026), IEF 6.6%
- ▣ Risk contributions grow with volatility: risk-based rules give Bitcoin a small weight automatically
- ▣ Robust mean–variance for crypto-assets: [Pele \(2023\)](#); Seminar 4, B5

Recap: BVB Portfolios

- Benchmark choice matters: BET omits about 7% a year of dividends
- No optimised BVB-only rule beats BET-TR significantly
- Eight stocks with average correlation 0.44: little room for diversification inside the BVB
- Open question: does diversification **beyond** BET pay? (Seminar 4, Part C)

An Open Question

- Does any **estimated** allocation rule beat $1/N$ out of sample, net of costs, once every rule, window and universe tried is counted?
- Live evidence points both ways
 - ▶ Nonlinear shrinkage of Σ improves GMV out of sample in large cross-sections: [Ledoit & Wolf \(2017\)](#)
 - ▶ Combining tangency, GMV and $1/N$ under estimation risk: [Kan, Wang & Zhou \(2022\)](#)
 - ▶ Backtests need haircuts for the number of specifications tried: [Harvey & Liu \(2015\)](#)
- Formal statement for K rules on one universe
 - ▶ H_0 : $\Delta_k = SR_k - SR_{1/N} \leq 0$ for all k ; H_1 : $\Delta_k > 0$ for some k
 - ▶ Statistic: $\hat{\Delta}_k$ with HAC or block-bootstrap errors ([Ledoit & Wolf \(2008\)](#)); error rate fixed for the whole family
- Falsified by: a positive $\hat{\Delta}_k$ that survives a family-wise correction, costs and a second sample

The Discovery Loop, Step by Step (1/2)

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature:** *"List peer-reviewed studies since 2015 comparing shrinkage or risk-based rules with $1/N$ out of sample; for each give DOI, N , T , cost assumption and test."*
 - ▶ Resolve every DOI on Crossref and open the table cited; assistants merge authors and invent DOIs
- **Hypotheses:** *"Give three competing explanations for a Sharpe gain over $1/N$ (better Σ , a low-volatility premium, luck) and the testable implication of each for turnover, subperiods and factor loadings."*
- **Data and code:** *"Write Python that reads data/market, joins prices on common dates and runs a 60-month rolling GMV-LW and $1/N$; weights at t may use data up to $t - 1$ only."*
 - ▶ Replicate first: the combined-universe table of this chapter ($1/N$ 0.76, GMV-LW 0.20) before changing anything

The Discovery Loop, Step by Step (2/2)

- **Robustness:** *"Draft a pre-registration: the family of tests ($\text{rules} \times \text{universes} \times \text{windows} \times \text{costs}$), the error rate (FWER or FDR), the subperiods and a placebo, fixed before any result is seen."*
 - ▶ Placebo: random long-only weights with the same turnover; a real effect must beat their distribution
- **Critique:** *"Act as a referee: give the three strongest reasons my Sharpe gain is spurious and the test that would expose each."*
- **Report:** every specification tried, adjusted p -values, the power of the design, code and seeds
- Fix the plan before the first run: any specification chosen after seeing results is exploratory and is reported as such

What the Human Must Check

- **Look-ahead:** weights at t built with month- t returns $\Rightarrow$ assert that the last estimation date is before t ; lag the weights one more month, the result must barely move
- **Survivorship and selection:** an ETF list chosen today $\Rightarrow$ use only assets listed at the start of each window
- **Multiple testing:** 7 rules $\times$ 3 universes already give 21 tests $\Rightarrow$ Holm (FWER) or Benjamini–Hochberg (FDR, [Benjamini & Hochberg \(1995\)](#)) adjusted p -values
- **Power:** $\hat{\Delta}$ has a standard error near 0.18 on 171 months $\Rightarrow$ compute the detectable effect before running
- **Return type:** portfolios aggregate **simple** returns; Sharpe ratios use **excess** returns $\Rightarrow$ recompute one month step by step
- **Invented references:** an unresolvable DOI or a title mismatch $\Rightarrow$ drop the claim, log it in `AI_ERRORS.md`

Mini-Case: Data Snooping in This Chapter's Backtest

- A typical assistant proposal: “MV-LO beats 1/N on the multi-asset universe”; $H_0: \Delta \leq 0$
- Replicate: $\hat{\Delta} = +0.29$, HAC standard error 0.18, bootstrap $p = 0.12$ (Section 7)
- Test the family, not the winner: 21 tests (7 rules $\times$ 3 universes)
 - ▶ Only 7 of 21 differences are positive; the 4 HAC p -values below 0.05 are all **losses** against 1/N
 - ▶ Smallest adjusted p -value: Holm 0.29, Benjamini–Hochberg 0.21; 0 discoveries at an FDR of 10%
 - ▶ MV-LO: Benjamini–Hochberg 0.31, Holm 1.00
- Power: detecting $\Delta = 0.29$ with power 0.8 at 5% needs about 552 months (46 years) out of sample
- Interpretation: the data cannot rank these rules; “no significant gain” is a bound on Δ , not proof of equality

Your Turn: A Team-Project Seed

- Question: does nonlinear shrinkage ([Ledoit & Wolf \(2017\)](#)) or a three-fund combination ([Kan, Wang & Zhou \(2022\)](#)) beat $1/N$ on the 16 ETFs of this chapter, after 10–50 bp costs?
- Replicate first: the combined-universe table (1/N Sharpe 0.76, MV-LO 0.81) and the adjusted p -values of the mini-case
- Deliverables
 - ▶ A one-page pre-registration: hypotheses, family of tests, error rate, power, placebo
 - ▶ Code on data/market, a table of all specifications, a referee-style self-critique
- Log every prompt, tool and accepted output in `AI_USE.md`; every error caught (invented DOI, look-ahead, wrong return type) and the test that caught it in `AI_ERRORS.md`
- Code of the mini-case: [Quantlets/Ch_04/ai_discovery_case.py](#)

Conclusions

- ▣ Markowitz is the right question with fragile answers: estimation error dominates
- ▣ Sample MV needs decades of data to beat $1/N$; out of sample it fails in every universe
- ▣ Shrinkage, constraints, Black–Litterman and risk-based rules make portfolios stable
- ▣ No rule beats $1/N$ significantly here; several lose after costs
- ▣ Evaluate out of sample, net of costs, with a test of the Sharpe difference

Links to Other Chapters

- ▣ **Chapter 3** (factor models): the tangency portfolio and the CAPM; factor covariance models
- ▣ **Chapters 5–6**: time-varying volatility and correlation for dynamic weights
- ▣ **Chapter 7**: portfolio Value-at-Risk and Expected Shortfall
- ▣ **Chapter 8**: backtesting without overfitting
- ▣ **Chapter 13**: machine learning for returns and portfolios

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Hierarchical Clustering
- ▣  Minimum Spanning Tree
- ▣  Multivariate Statistical Analysis

Code and data

- ▣ All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_04
- ▣ Risk-free rate: Kenneth French Data Library

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